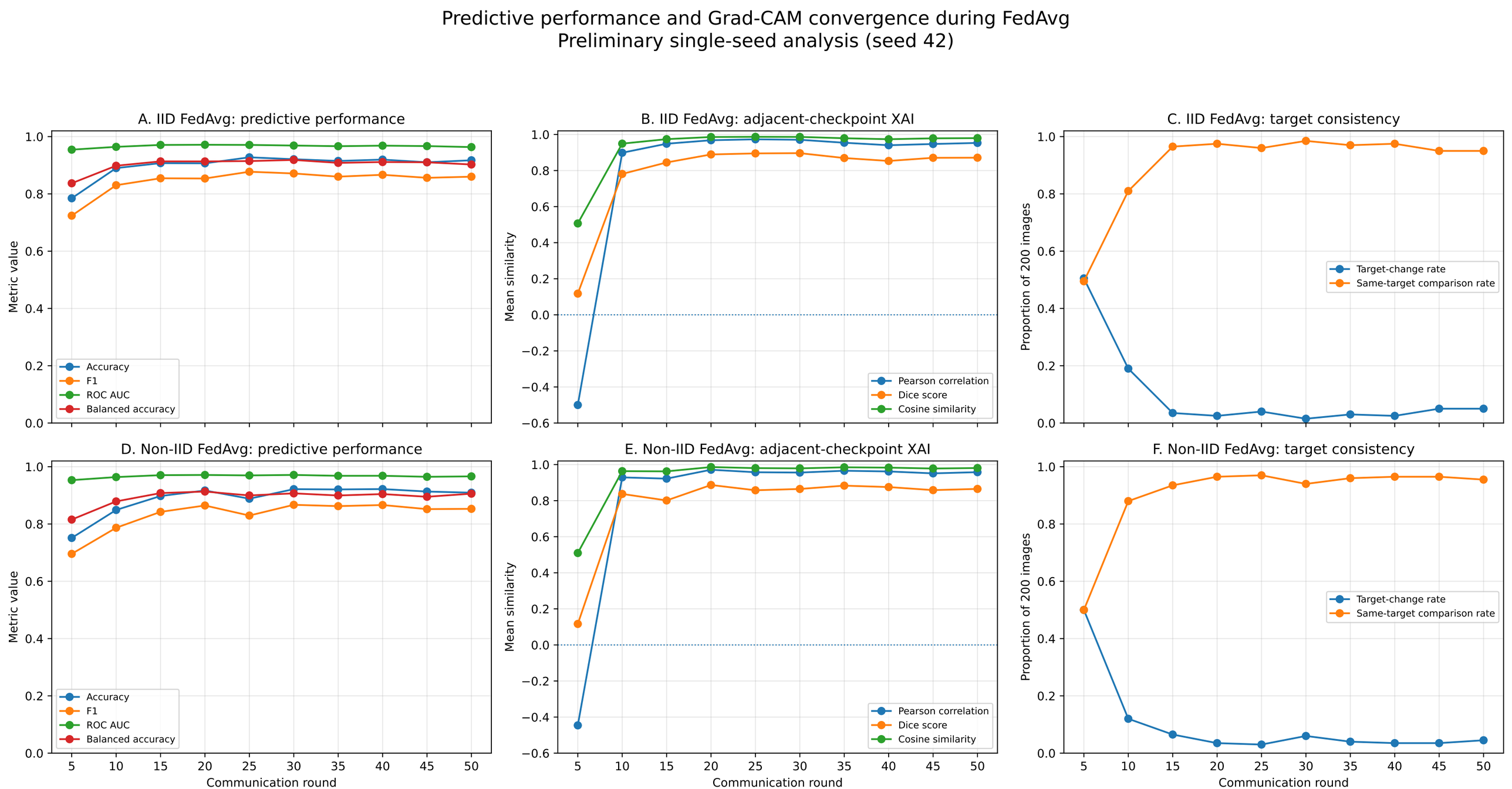
Predictive performance and Grad-CAM convergence during FedAvg
Preliminary single-seed analysis (seed 42)
A. IID FedAvg: predictive performance
B. IID FedAvg: adjacent-checkpoint XAI
C. IID FedAvg: target consistency
D. Non-IID FedAvg: predictive performance
E. Non-IID FedAvg: adjacent-checkpoint XAI
F. Non-IID FedAvg: target consistency
Metric value
Mean similarity
Proportion of 200 images
Communication round
Accuracy
F1
ROC AUC
Balanced accuracy
Pearson correlation
Dice score
Cosine similarity
Target-change rate
Same-target comparison rate
Predictive metrics use all 3,000 validation images. Grad-CAM similarity means use only same-target comparisons between each checkpoint and the preceding saved checkpoint.